Year 13 Further Maths - Lines and planes in space

Similar questions - Distances, reflections and modelling

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The plane Π has equation $2x - y + 2z = 6$. The point A has coordinates $(5,1,3)$. Find the coordinates of the point B, the reflection of A in Π. **[6]**

2 The non-intersecting lines l_1 and l_2 have equations $l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$. Find the exact shortest distance between l_1 and l_2. **[5]**

3 A and B have coordinates $(1,0,2)$ and $(3,4,8)$. Plane Π_A contains A and plane Π_B contains B. The planes are parallel, the x and y components of any normal to Π_A are equal, and the shortest distance between Π_A and Π_B is 2. There are two possible planes for Π_A. Determine the acute angle between these two possible planes. **[8]**

4 A construction site has four vertical poles whose bases are on the floor $z = 0$ and whose tops are at $P_1(0,0,12)$, $P_2(8,6,14)$, $P_3(0,10,5)$ and $P_4(12,2,9)$ (units are metres). A wire S runs from the top of P_1 to the top of P_2 and has vector equation $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 12 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix}$, $0 \leqslant \lambda \leqslant 2$. A wire T runs from the top of P_3 to the top of P_4.

(a) Find a vector equation of the line segment representing T, with an integer direction vector whose only positive common factor is 1. **[2]**

(b) Test 1: the wires must be at least 5 m apart at all positions. Determine whether the site passes Test 1. **[4]**

(c) Test 2: a camera at a point Q on S must be at least 19 m from the top of P_3. Determine whether it is possible to choose Q so that the site passes Test 2. **[3]**

Question 1

$\mathbf{n} = (2,-1,2)$, $|\mathbf{n}|^2 = 9$. $\mathbf{n}\cdot\mathbf{a} = 10-1+6 = 15$, so A is $\dfrac{15-6}{9} = 1$ unit of $\mathbf{n}$ from the plane (in the direction of $\mathbf{n}$). The foot of the perpendicular is $A-\mathbf{n} = (3,2,1)$ (check: $6-2+2=6$ ✓) and $B = A-2\mathbf{n} = (1,3,-1)$

Question 2

$\mathbf{d}_1\times\mathbf{d}_2 = (-4,3,5)$, magnitude $\sqrt{50} = 5\sqrt{2}$. $\mathbf{a}_2-\mathbf{a}_1 = (-1,-2,3)$, and its dot product with $(-4,3,5)$ is $4-6+15 = 13$. Distance $= \dfrac{13}{5\sqrt{2}} = \dfrac{13\sqrt{2}}{10}$

Question 3

Let $\mathbf{n} = (p,p,q)$. $\overrightarrow{AB} = (2,4,6)$ so $\overrightarrow{AB}\cdot\mathbf{n} = 6p+6q$. The distance is $\dfrac{|6p+6q|}{\sqrt{2p^2+q^2}} = 2$, so $36(p+q)^2 = 4(2p^2+q^2)$, giving $7p^2+18pq+8q^2 = 0$, i.e. $(7p+4q)(p+2q) = 0$. Normals: $\mathbf{n}_1 = (-2,-2,1)$ (from $p=-2q$) and $\mathbf{n}_2 = (-4,-4,7)$ (from $7p=-4q$). $\cos\theta = \dfrac{|8+8+7|}{3\times 9} = \dfrac{23}{27}$, so $\theta = 31.6°$

Question 4

(a) $\overrightarrow{P_3P_4} = (12,-8,4) = 4(3,-2,1)$. $\mathbf{r} = \begin{pmatrix}0\\10\\5\end{pmatrix} + \mu\begin{pmatrix}3\\-2\\1\end{pmatrix}$, $0\leqslant\mu\leqslant 4$

(b) $\mathbf{d}_S\times\mathbf{d}_T = (4,3,1)\times(3,-2,1) = (5,-1,-17)$, magnitude $\sqrt{315}$. $\mathbf{a}_T-\mathbf{a}_S = (0,10,-7)$, dot product $-10+119 = 109$. Distance between the lines $= \dfrac{109}{\sqrt{315}} = 6.14$ m. The closest points occur at $\lambda = 1.62$ and $\mu = 2.74$, both within the allowed ranges, so the shortest distance between the wires is $6.14 \geqslant 5$: the site passes Test 1

(c) $Q = (4\lambda, 3\lambda, 12+\lambda)$. $|QP_3|^2 = 16\lambda^2+(3\lambda-10)^2+(7+\lambda)^2 = 26\lambda^2-46\lambda+149$. This is a quadratic with positive coefficient of λ^2, so its maximum on $0\leqslant\lambda\leqslant 2$ is at an end: 149 at $\lambda = 0$ and 161 at $\lambda = 2$. The largest distance is $\sqrt{161} = 12.7 < 19$, so it is not possible to meet Test 2